

Question	Answer	Marks
9(a)(i)	electromagnetic wave can behave like a particle	B1
	moving particle can behave like a wave	B1
9(a)(ii)	$\lambda = h / p$	M1
	h is the Planck constant	A1
9(b)(i)	<i>Any two points from:</i> • pattern similar to diffraction of light • diffraction (pattern) is characteristic of wave behaviour • rings show constructive and destructive interference	B2
9(b)(ii)	(de Broglie) wavelength decreases	B1
	rings become closer together	B1

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